

Loops and Stopping

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Trace Tables

Without opening the old lesson, explain one key idea from Trace Tables. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Decompose a Goal

Without opening the old lesson, explain one key idea from Decompose a Goal. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace before running for Loops and trace tables. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Loops and trace tables. Proposed explanation: “The stop value is included”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Changed independent case

Start count at 1. While count < 8, add 3. Trace the condition and values. What happens if the condition becomes count != 8?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Trace $\text{count}=0$; while $\text{count}<4$: $\text{count}=\text{count}+2$. List values tested by the condition and final count. Would replacing <4 with $\text{count}=5$ terminate?

6. Solve the given case

Start $n=0$; while $n<3$, add 1 and record n .

List records and stopping condition.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Start at count 0. While count is less than 3, add 1 and display count. List the displays and the stopping value.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 5 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Loop through every candidate book, update its score, then stop at the end of the list.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Trace `count=0; while count<4: count=count+2`. List values tested by the condition and final count. Would replacing `<4` with `!=5` terminate?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Explain the range

List the values supplied by `range(2, 5)` and explain why 5 is absent. How many times does its body run?

My reasoning and evidence

13. W2 Trace before running

Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each iteration. Predict the single output, then run and compare.

My reasoning and evidence

14. W3 Repair the reset

For the faulty script with range(1, 4), record total after each addition. Explain the wrong final output and remove only the faulty reset.

My reasoning and evidence

15. W4 Build a repeated addition independently

A box starts with 5 counters. Each round adds 2 counters. Use rounds = 3, an initialized total and for turn in range(rounds). Print the final total once after the loop. Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

My reasoning and evidence

16. W5 Check a contract limitation

In your counter script, range(-2) performs no iterations. Does returning the initial 5 make negative rounds valid? Explain what the script checks and what it assumes.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Loops and Stopping

A fictional counter starts at 12 and decreases while above a boundary.

Python: `count=12; while count>3: count=count-4; print` after the loop.

The condition is checked before each subtraction.

Loops and Stopping

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Initial check

$12 > 3$: true; subtract 4 to get 8.



2. Next check

$8 > 3$: true; subtract 4 to get 4.



3. Final update

$4 > 3$: true; subtract 4 to get 0.



4. Stop

$0 > 3$ is false; report 0 after three updates.

CHANGE ONE THING

Replace $\text{count} > 3$ with $\text{count} \leq 3$.

Explain the changed result using the matching lesson.

Why is report 0 correct for the original loop, and why is equality-only stopping risky here?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Use progress and stopping arguments for any repeated recommender process; test inputs that require zero, one and several updates.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

Use progress and stopping arguments for any repeated recommender process; test inputs that require zero, one and several updates.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional counter starts at 12 and decreases while above a boundary.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? $0 > 3$ is false; report 0 after three updates.

1. Next check
2. Final update
3. Stop
4. Initial check

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Stop $0 > 3$ is false; report 0 after three updates. The condition does not promise the counter ends exactly at 3. Subtractions can cross the boundary. The count strictly decreases, so the below-or-equal stopping region is eventually reached from this start. A condition requiring exact equality can be unsafe when repeated updates skip that value.

Practice 2: explain the deciding evidence

Changed-case practice: Replace $\text{count} > 3$ with $\text{count} \neq 3$. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. From 12, the loop never reaches 3 and does not terminate in this toy model.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: From 12, the loop never reaches 3 and does not terminate in this toy model. 12,8,4,0,−4,... are multiples of 4; none equals 3.

Repair a missing building block

Review Trace Tables (ai-2/4). Goal: Trace variable values before and after each loop iteration. Short practice: Move print inside the original loop. What sequence is displayed?. Return to this week and explain what you changed.

Review Decompose a Goal (ai-2/2). Goal: Distinguish a function definition from a function call. Short practice: What happens visibly if the caller stores needed but omits print(needed)?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
